



Informatica® Intelligent Cloud Services
February 2024

Model Serve

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Preface

Use *Model Serve* to learn how to register, deploy, and monitor machine learning models in Model Serve.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit <https://docs.informatica.com>.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

Informatica Intelligent Cloud Services web site

You can access the Informatica Intelligent Cloud Services web site at <http://www.informatica.com/cloud>. This site contains information about Informatica Cloud integration services.

Informatica Intelligent Cloud Services Communities

Use the Informatica Intelligent Cloud Services Community to discuss and resolve technical issues. You can also find technical tips, documentation updates, and answers to frequently asked questions.

Access the Informatica Intelligent Cloud Services Community at:

<https://network.informatica.com/community/informatica-network/products/cloud-integration>

Developers can learn more and share tips at the Cloud Developer community:

<https://network.informatica.com/community/informatica-network/products/cloud-integration/cloud-developers>

Informatica Intelligent Cloud Services Marketplace

Visit the Informatica Marketplace to try and buy Data Integration Connectors, templates, and mapplets:

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Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit <https://search.informatica.com>. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

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The Informatica Intelligent Cloud Services Trust Center provides information about Informatica security policies and real-time system availability.

You can access the trust center at <https://www.informatica.com/trust-center.html>.

Subscribe to the Informatica Intelligent Cloud Services Trust Center to receive upgrade, maintenance, and incident notifications. The [Informatica Intelligent Cloud Services Status](#) page displays the production status of all the Informatica cloud products. All maintenance updates are posted to this page, and during an outage, it will have the most current information. To ensure you are notified of updates and outages, you can subscribe to receive updates for a single component or all Informatica Intelligent Cloud Services components. Subscribing to all components is the best way to be certain you never miss an update.

To subscribe, on the [Informatica Intelligent Cloud Services Status](#) page, click **SUBSCRIBE TO UPDATES**. You can choose to receive notifications sent as emails, SMS text messages, webhooks, RSS feeds, or any combination of the four.

Informatica Global Customer Support

You can contact a Global Support Center through the Informatica Network or by telephone.

To find online support resources on the Informatica Network, click **Contact Support** in the Informatica Intelligent Cloud Services Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at <https://www.informatica.com/services-and-training/support-services/contact-us.html>.

CHAPTER 1

Introducing Informatica Model Serve

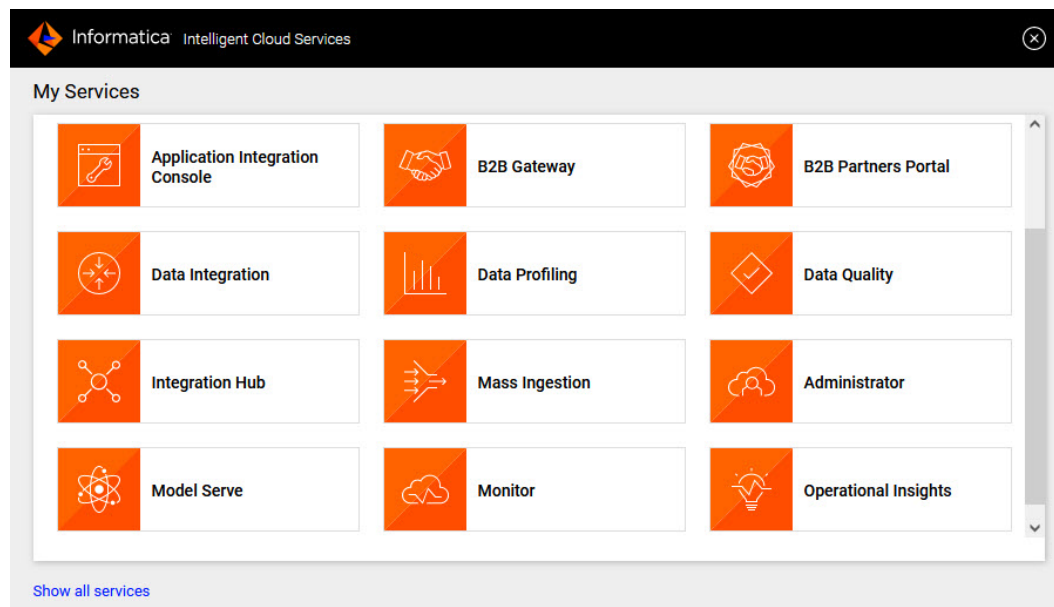
Welcome to Informatica Model Serve! Use Model Serve to deploy machine learning models so you can generate predictions for your business based on historical data. You can deploy a quick start model to get started with a standard model, or you can upload your own files and deploy a user-defined model.

Model Serve is one of multiple services available in Informatica Intelligent Cloud Services. When you log in to Informatica Intelligent Cloud Services, the **My Services** page opens onscreen. In addition to Model Serve, the **My Services** page might include other services that you subscribe to and trial versions of other services.

As a Model Serve user, you might work with the following services:

- Model Serve. Register and deploy machine learning models, and monitor deployments.
- Administrator. Perform administrative tasks, such as managing privileges.

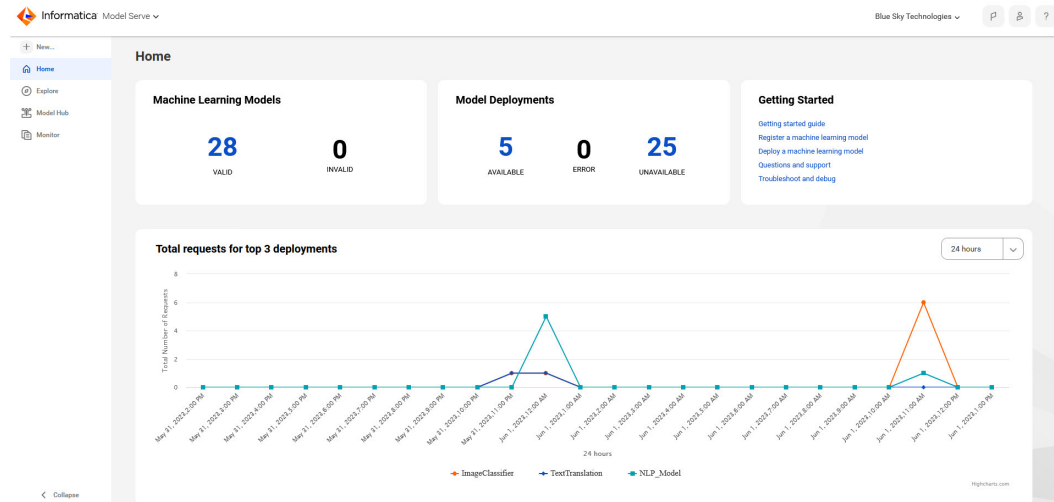
The following image shows an example **My Services** page:



Home page

When you select Model Serve from the **My Services** page, the Model Serve **Home** page appears.

The following image shows the **Home** page:



The page displays the following panels:

- Machine Learning Models. The number of machine learning models in your organization.
- Model Deployments. The number of model deployments in your organization that appear on the **Monitor** page.
- Getting Started. Links to documentation and support.
- Total requests for top 3 deployments. The number of prediction requests for the most-used model deployments.

Note: The chart uses the time zone configured in your user profile.

You can access the following pages from the navigation bar:

- New. Create a new machine learning model or model deployment.
- Home. Return to the **Home** page.
- Explore. Create and manage projects and assets.
- Model Hub. View and deploy quick start models.
- Monitor. View and manage the status of model deployments.

Asset types

Use assets in Model Serve to register and deploy machine learning algorithms.

You can use the following types of assets:

Quick start model

A quick start model is a machine learning algorithm that Informatica makes ready to deploy. Use quick start models to quickly deploy an algorithm for common use cases.

Machine learning model

A machine learning model contains the artifacts for a user-defined algorithm that you trained using machine learning techniques to categorize input data. Use a machine learning model when you want to deploy your own algorithm.

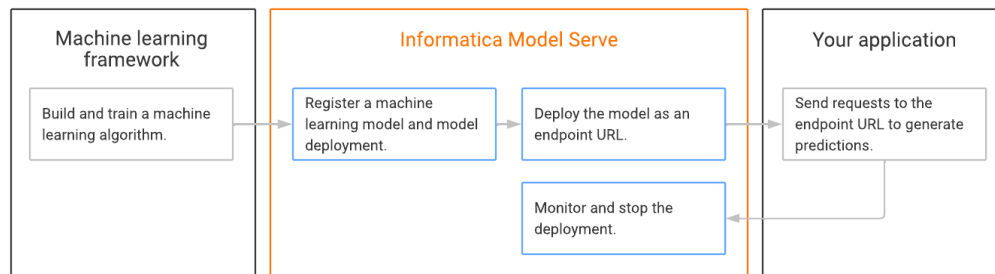
Model deployment

A model deployment defines the runtime configuration to deploy a machine learning model, such as the maximum number of compute units that the deployment can use.

Deployment process

Deploying a model in Model Serve involves tasks that you perform in multiple applications. The process differs based on the type of model that you use.

The following image shows the process to deploy a model:



The following table describes the tasks in the deployment process:

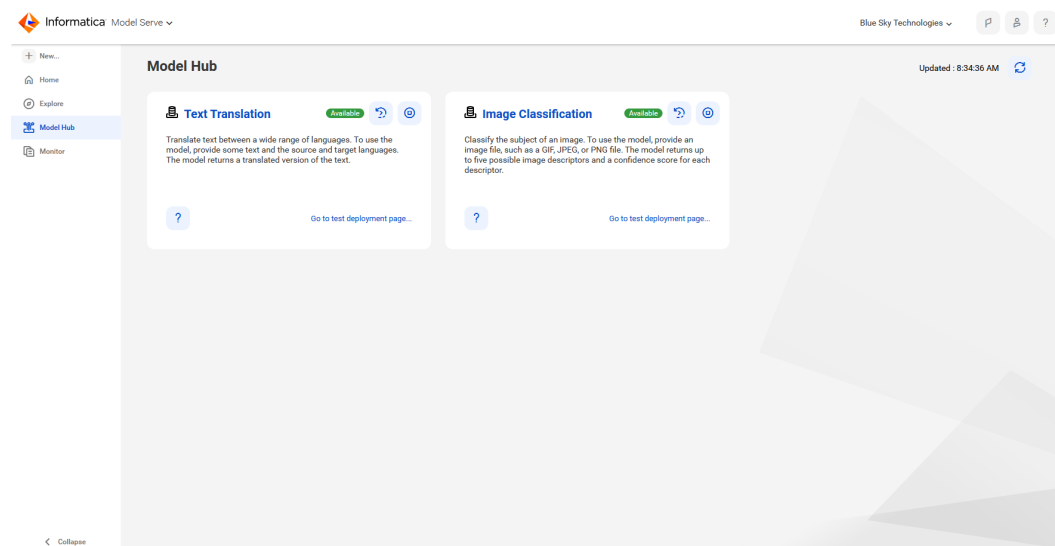
Task	Required for	Description
Build and train a machine learning algorithm.	User-defined models	Use a third-party machine learning framework, such as TensorFlow, to build and train an algorithm.
Register a machine learning model and a model deployment.	User-defined models	In Model Serve, configure a machine learning model where you upload the algorithm files. Then create a model deployment to define the runtime properties for deploying the model.
Deploy the model as an endpoint URL.	Quick start models and user-defined models	Start the quick start model or model deployment to make the endpoint URL available for requests.
Send requests to the endpoint to generate predictions.	Quick start models and user-defined models	In your application, send API calls to the endpoint URL to request predictions from the deployed model.
Monitor and stop the deployment.	Quick start models and user-defined models	Use Model Serve to monitor the status of your deployments, download logs for troubleshooting, and stop deployments to release the cloud resources.

CHAPTER 2

Quick start models

Use a quick start model to quickly deploy a machine learning algorithm that is already trained and registered. You can access quick start models on the **Model Hub** page.

The following image shows the **Model Hub** page:



When a quick start model is available, you can click on the name of the model on the **Model Hub** page or the **Monitor** page to view information about the model and test it. You can view the status of the quick start models on the **Model Hub** page and the **Monitor** page.

Image classification model

The image classification model generates images descriptors to classify the subject of an image.

The model deploys the [ResNet-50 v1.5 model from Hugging Face](#). It requires a JPEG, JPG, or PNG image file that is a maximum of 5 MB as input. The model returns five classifications with a confidence score for each classification.

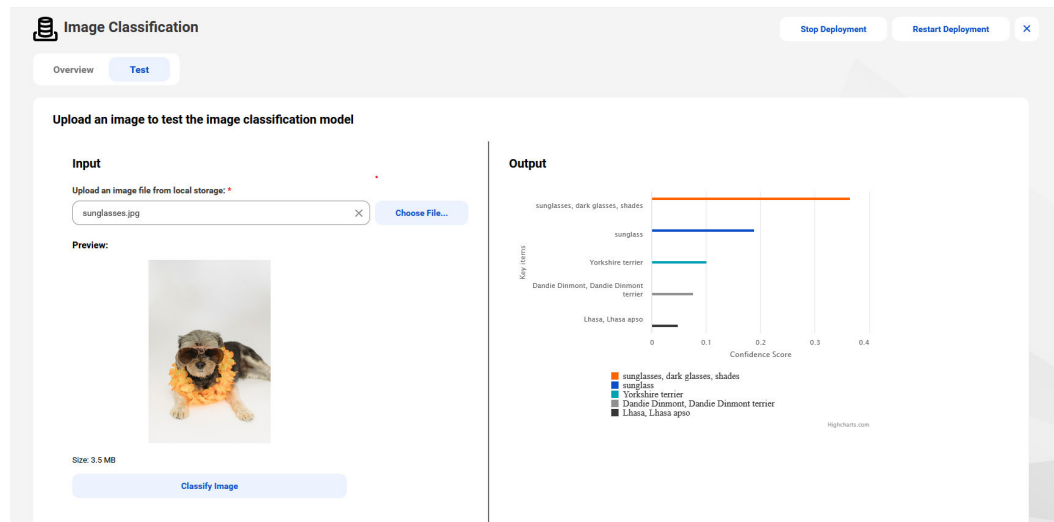
The results from the image classification model might differ from the results you see on Hugging Face for the ResNet-50 v1.5 model. Hugging Face sometimes changes the model on their website, while the image classification model is static.

Note: This model is sourced from a third party. Informatica does not assume ownership of its behavior, functionality, or any associated liabilities. Any issues, updates, or concerns related to this artifact should be directed to the original provider or creator. We provide this artifact as-is and without warranties.

Example

You run an online store that receives customer photos in reviews. Before the photos are posted to your website, you need to check that they're relevant to the product being reviewed. As an initial check, you use the image classification model to classify the photos. If a photo returns the right product type with a confidence score over 0.5, you can publish the review. Otherwise, you flag the photo to be reviewed by a content moderator.

The following image shows an example test of the image classification model:



Text translation model

The text translation model translates text between a wide range of languages.

The model deploys the [M2M100 418M model from Hugging Face](#). The model requires the source language code, target language code, and source text, and it returns the translated text. The source text can be a maximum of 3,000 characters. To see the list of languages that the model supports and the language codes, refer to the Hugging Face documentation.

The results from the text translation model might differ from the results you see on Hugging Face for the M2M100 418M model. Hugging Face sometimes changes the model on their website, while the text translation model is static.

Note: This model is sourced from a third party. Informatica does not assume ownership of its behavior, functionality, or any associated liabilities. Any issues, updates, or concerns related to this artifact should be directed to the original provider or creator. We provide this artifact as-is and without warranties.

Example

You're designing the website for a local government. In order to make the information accessible to speakers of different languages, you use the text translation model to render the website in the language configured in the user's browser.

The following image shows an example test of the text translation model:

The screenshot shows the 'Text Translation' model interface. At the top, there are tabs for 'Overview' and 'Test', with 'Test' being the active tab. Below the tabs, there's a section titled 'Enter text to test the Text Translation model'. This section is divided into 'Input' and 'Output' columns. In the 'Input' column, there are dropdown menus for 'Source Language' (set to 'English') and 'Target Language' (set to 'Spanish'). Below these is a text area for 'Text for Translation' containing the text: 'Redwood City boasts over 30 unique parks from small neighborhood parks that are only a quarter of an acre, to larger multiuse parks that are 40 acres, to Off-leash Dog Parks and a Skate Park. In addition, Redwood City has two outdoor pools that are operated seasonally during the Summer.' At the bottom of the input section are 'Translate' and 'Reset' buttons. The 'Output' column shows the translated text in Spanish: 'Redwood City cuenta con más de 30 parques únicos, desde pequeños parques vecinales que son sólo un cuarto de acre, hasta parques de uso múltiple más grandes que son 40 acres, hasta parques de perros off-leash y un Skate Park. Además, Redwood City tiene dos piscinas al aire libre que se operan estacionalmente durante el verano.'

Deploying a quick start model

To use a quick start model, deploy it from the **Model Hub** page and make requests through the API.

1. On the **Model Hub** page, find the desired quick start model and click **Start Deployment**.
2. When the status becomes *Available*, click the name of the quick start model on either the **Model Hub** page or the **Monitor** page. The overview page opens and you can view information about the deployed quick start model, including the model endpoint URL.

The following image shows the overview information:

The screenshot shows the 'Text Translation' model overview page. The page has a sidebar on the left with navigation links: 'New...', 'Home', 'Explore', 'Model Hub', 'Monitor', and 'Text Translation' (which is highlighted). The main content area is titled 'Text Translation' and has tabs for 'Overview' and 'Test', with 'Overview' being the active tab. Below the tabs, there's a table with the following data:

Status	Requests	Duration	Max compute units	Start time	Started by
Available	1	19 hours 56 minutes	4	Sep 20, 2023, 12:59 PM	evelina

Below the table, there's a section titled 'Model endpoint URL' with the text: 'REST API endpoint to communicate with the machine learning model.' and the URL: 'https://[redacted]/ml-predict/api/v1/deployment/[redacted]'.

3. Use the endpoint URL to make prediction requests through the API. For more information about making API requests, see [Model Serve REST API](#).
4. To release the resources, stop the deployment when you no longer need it.

CHAPTER 3

User-defined models

A user-defined model is a machine learning model that you build and train outside of Model Serve and upload to Model Serve to deploy it. To deploy a user-defined model, register a machine learning model and then create and start a model deployment.

Registering a machine learning model

To get started with Model Serve, register a machine learning model based on a trained machine learning algorithm.

When you register a machine learning model, Model Serve uploads the model files to an internal file store. After you register a model, you can use it in one or more model deployments.

To configure a machine learning model, perform the following steps:

1. Define general properties for the model.
2. Upload the model artifacts as a ZIP file.
3. Define the input fields that the model expects and the output fields that the model returns.

After you save the machine learning model, the overview page appears. Users in your organization can download the model artifacts from the overview page.

If you need to edit a machine learning model, stop all model deployments that use the model. You can't edit a machine learning model if a running model deployment uses it.

Define a machine learning model

Create a machine learning model and configure the general properties.

1. In the navigation bar, select **New > Machine Learning Model**.

2. Configure general properties for the model. The following tables describes the properties to configure:

Property	Description
Model name	Required. Name of the machine learning model. The name must begin with a letter and can contain alphanumeric characters, hyphens (-), and underscores (_). Maximum length is 100 characters.
Description	Description of the machine learning model.
Location	Required. Project folder in which the machine learning model resides. If the Explore page is currently active and a project or folder is selected, the default location for the machine learning model is the selected project or folder. Otherwise, the default location is the <i>Default</i> folder.

Upload model artifacts

Download and edit the code template, and then upload the model artifacts.

1. Select *Custom* as the framework.
2. Download and unzip the code template.
The code template includes the files `predict.py` and `requirements.txt`.
3. Edit the `predict.py` file to define the entry point into your custom algorithm files. Define the `__init__()` method to load your model files when the model deployment starts. Define the `predict()` method to take a JSON-formatted request and return a prediction from the model.

Note: Defining the `__init__()` method isn't required. If you don't define the method, the model deployment loads the model files for each request, which slows down the performance of the deployment.

4. Edit the `requirements.txt` file to list the third-party libraries required to run your machine learning algorithm.

For example, the file might contain the following list:

```
scikit-learn
scipy
tensorflow
numpy==1.19.5
```

Note: If you don't specify a version for a library, Model Serve uses the newest version. Ensure that the library versions you use are compatible with KServe 0.8.0. If the library version that's compatible with KServe 0.8.0 isn't compatible with another library that your model needs, contact Informatica Global Customer Support to upgrade your version of KServe.

5. Compress `predict.py`, `requirements.txt`, and your algorithm files into a ZIP file. The file can be up to 3 GB. The `predict.py` and `requirements.txt` files must be at the root level in the ZIP file.
6. Upload the ZIP file as the model file.

Define input and output fields

Define the input fields that the model expects in API requests and output fields that the model returns in API responses.

On the **Fields** tab, add the name and data type of each input and output field. Configure at least one input field and output field.

The following image shows the **Fields** tab:

The screenshot shows the 'New Machine Learning Model' interface with the 'Fields' tab selected. The interface has a top bar with a logo, the title 'New Machine Learning Model', and navigation buttons: '< Back', 'Next >', 'Save', 'Cancel', and a close button 'X'. Below the top bar is a tab bar with three tabs: '1 General', '2 Model Artifacts', and '3 Fields'. The 'Fields' tab is active. The main content area is titled 'Specify the Input and Output fields for your model' and shows 'File Name : MNIST.zip'. There are two sections: 'Input Fields' and 'Output Fields'. Each section has a dropdown arrow on the right. The 'Input Fields' section has three input fields: 'Name: *' with the value 'pixels', 'Data Type: *' with a dropdown menu showing 'Array of Float', and 'Shape: *' with a dropdown menu showing '[28,28]'. There is an 'Add Field' button below these fields. The 'Output Fields' section has three input fields: 'Name: *' with the value 'classify', 'Data Type: *' with a dropdown menu showing 'Array of Float', and 'Shape: *' with a dropdown menu showing '[1,10]'. There is an 'Add Field' button below these fields.

Note: When you output a user-defined object, ensure that the `predict()` method returns a serialized JSON string, and then configure a string output field.

Deploying a machine learning model

Deploy a machine learning model to expose an endpoint URL for API requests.

To deploy a machine learning model, first configure a model deployment that defines how you want the deployment to run. When you start the model deployment, Model Serve provisions the required cloud resources and deploys the machine learning model. After the model deployment becomes available, you can make requests to the endpoint URL.

To deploy a machine learning model, perform the following tasks:

1. Configure runtime properties in a model deployment.
2. Start the model deployment.
3. Optionally, test that the model is generating predictions as you expect.

Configure a model deployment

Configure a model deployment to define the runtime properties used to deploy a machine learning model.

1. Select **New > Model Deployment**.

2. On the General tab, configure general properties for the model deployment. The following table describes the properties to configure:

Property	Description
Model deployment name	Required. Name of the model deployment. The name must begin with a letter and can contain alphanumeric characters, hyphens (-), and underscores (_). Maximum length is 100 characters.
Description	Description of the model deployment.
Location	Required. Project folder in which the model deployment resides. If the Explore page is currently active and a project or folder is selected, the default location for the model deployment is the selected project or folder. Otherwise, the <i>Default</i> folder is the default location.
Machine learning model	Required. Machine learning model to deploy.

3. On the Compute tab, set the **Compute Units** property to the maximum number of compute units that you want the model deployment to use.

One compute unit includes 4 CPUs and 16 GB of memory. The default is 4 compute units.

4. Save the model deployment.

After you configure and save the model deployment, you can start the deployment or close the editor to view the model deployment overview page.

Note: You can't edit a model deployment while it's running. To edit, stop the model deployment and wait until it's unavailable.

Start the model deployment

When you start a model deployment, Model Serve deploys your machine learning model and makes the URL endpoint available for requests.

You can start a model deployment in any of the following ways:

- In the model deployment editor, select **Start Deployment** after you save a valid deployment.
- On the model deployment overview page, select **Start Deployment**.
- On the **Monitor** page, open the **Actions** menu for a model deployment and select **Start**.

If there are no other deployments available when you start a model deployment, Model Serve brings up the necessary cloud resources, which can take up to 10 minutes.

When you request predictions from the model deployment, a request can run for up to five minutes. If the model doesn't return a response after five minutes, the request times out.

Test the model deployment

Send a request to verify that your model is generating predictions as you expect. You can generate test predictions for model deployments that are available.

1. Navigate to the **Test** tab in the model deployment overview.
2. Enter a request call in JSON format. Ensure that your call includes all of the necessary input fields.

3. Select **Send Request**.

Model Serve sends the request and then displays the response from the model. Model Serve returns the response in JSON format if possible. Otherwise, it returns the response as a String.

CHAPTER 4

Monitoring deployments

Use the **Monitor** page to check and manage the status of deployments and to download logs. The **Monitor** page includes both quick start models and user-defined deployments.

The following table describes the properties that the **Monitor** page displays for each deployment:

Property	Description
Name	Name of the deployment.
Status	Status of the deployment. A deployment can have one of the following statuses: - Starting. Model Serve is preparing to deploy the machine learning model. - Provisioning resources. Model Serve is provisioning cloud resources. - Deploying. Model Serve is deploying the model. - Available. The deployment started successfully and the endpoint URL is available for requests. - Stopping. The deployment is stopping. - Unavailable. The deployment is stopped and all resources are released. - Restarting. Model Serve is stopping and restarting the deployment. - Redeploying. Model Serve is redeploying the model after restarting. - Error. The deployment couldn't start because it encountered an error. If Model Serve provisioned cloud resources before it encountered the error, the resources might still be running while the status is Error. - Terminated. Model Serve terminated the deployment and released the cloud resources after the deployment encountered an error.
Location	Project folder where the model deployment resides. Not applicable to quick start models.
Started by	Name of the user that started the deployment.
Start time	Date and time that the deployment last reached the <i>Deploying</i> status. Note: If you start a deployment and it encounters an error while in the <i>Starting</i> or <i>Provisioning resources</i> status, the start time does not appear.
End time	Date and time that the deployment stopped.
Duration	Amount of time that the deployment has been running.
Requests	Total number of prediction requests that the deployment has received.

To ensure that the information on this page is current, Model Serve polls the Informatica Intelligent Cloud Services repository regularly and refreshes the monitoring details while you are active on the **Monitor** page.

If a user starts a deployment that the **Monitor** page didn't previously list, the new deployment doesn't appear on the page. The status message at the top of the page indicates that the information isn't up-to-date and prompts you to refresh the page to see new deployments.

Download logs

You can download log files to help you troubleshoot model deployments. You can download log files for a deployment that is starting, available, stopped, or failed.

1. Select the **Actions** menu, and then choose **Download Logs**.
2. In the **Download Logs** dialog box, set the time range for the logs you want to download. You can download logs created in the past 30 days.

CHAPTER 5

Project and asset management

You can manage projects and assets on the **Explore** page.

You can manage your Informatica Intelligent Cloud Services projects and assets in the following ways:

- Copy projects, folders, or assets to create new versions of them.
- Move folders or assets to other locations.
- Rename projects, folders, or assets.
- Delete projects, folders, or assets.
- Apply tags so you can filter for related assets on the **Explore** page.
- Configure permissions for projects, folders, or assets.
- Migrate assets between organizations.

Copying projects, folders, and assets

You can copy projects, folders, and assets on the **Explore** page. You might want to copy an object to use as a template, or you might want to create a backup copy.

When you copy objects, Informatica Intelligent Cloud Services retains child elements in the following instances:

- When you copy a project, the new project contains all of the folders and assets that were in the original project. Similarly, when you copy a folder, the new folder contains all of the assets that were in the original folder.
- When you copy assets or a folder containing assets to another location, the assets retain their references to dependent objects.
For example, the Development project contains the assets `primary_asset` and `reference_asset`, and `primary_asset` requires `reference_asset`. If you copy `primary_asset` to the Production project, the new `primary_asset` in the Production project references `reference_asset` in the Development project.

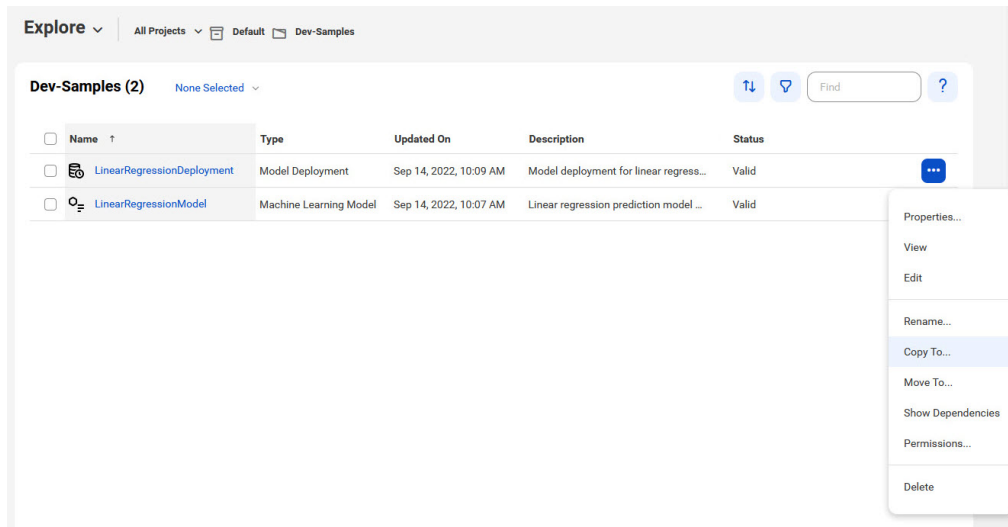
When you copy an asset within a folder, you can keep both assets. When you copy an asset into a different folder that contains an asset with the same name, you can overwrite the asset in the folder or keep both. If you choose to keep both assets, Informatica Intelligent Cloud Services appends the new asset name with "Copy x" where x is the sequential copy number.

Note: To avoid naming conflicts with duplicate assets, rename assets with a "Copy x" suffix.

Perform the following steps to copy an object:

1. On the **Explore** page, navigate to the object that you want to copy.

2. In the row that contains the object that you want to copy, click **Actions** and select **Copy To**.

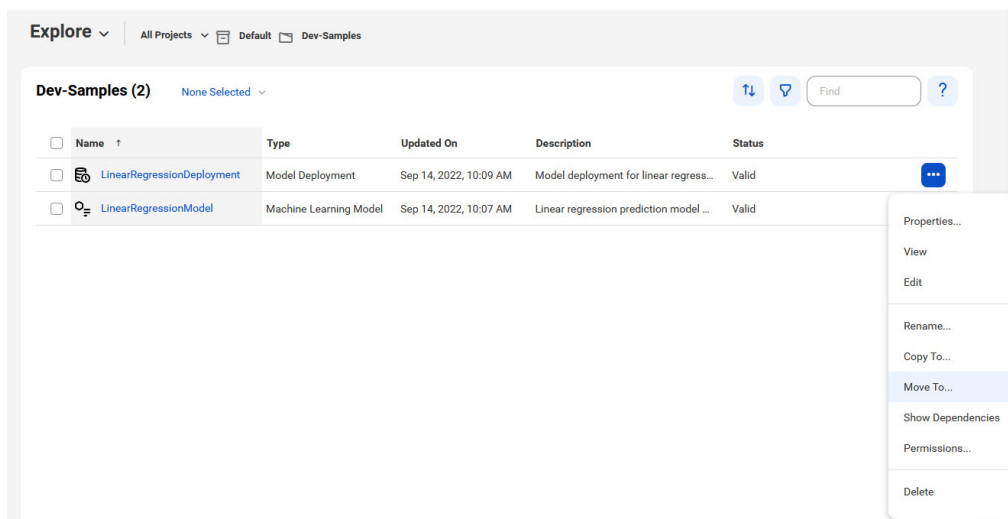


3. Browse to the new location and click **Select**.

Moving folders and assets

You can move folders and assets on the **Explore** page.

1. On the **Explore** page, navigate to the folder or assets that you want to move.
2. To move a folder or a single asset, in the row that contains the folder or asset, click **Actions** and select **Move To**, and then browse to the new location and click **Select**.



Renaming projects, folders, and assets

You can rename projects, folders, and assets without losing references to objects that they use.

For example, if an asset named `primary_asset` uses an asset named `reference_asset`, you can rename `primary_asset` to `main_asset` and it will still reference `reference_asset`.

You can't use the following characters in project, folder, asset, or tag names:

? ' | { } " ^ & [] / \

1. On the **Explore** page, navigate to the object that you want to rename.
2. To rename an asset, in the row that contains the asset, click **Actions** and select **Rename**.
3. To rename a project or folder, in the row that contains the project or folder, click **Actions** and select **Properties**.
4. Enter the new name and click **Save**.

You cannot use special characters in a name or use the same name as another object that is in the same folder.

Deleting projects, folders, and assets

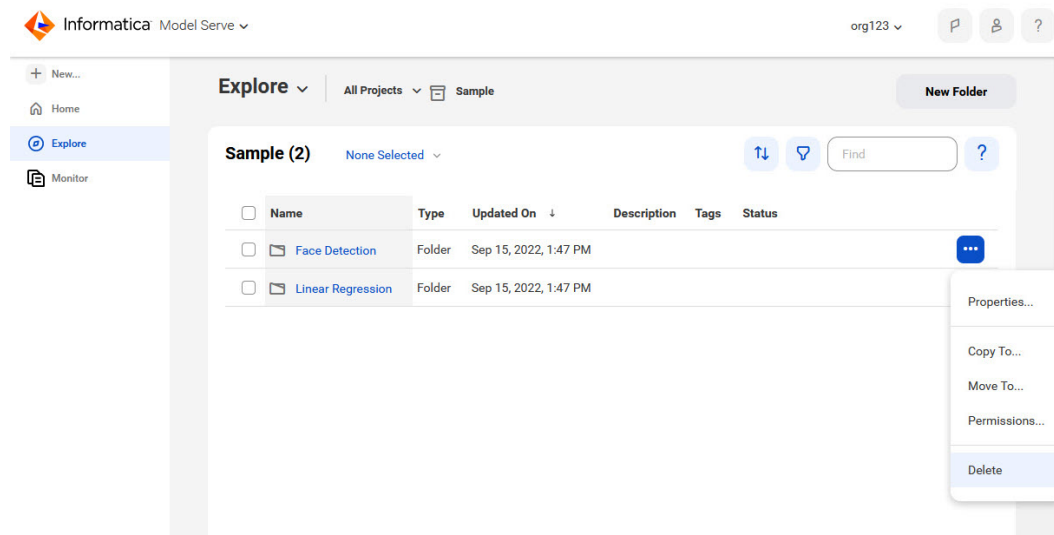
You can delete a project, folder, or asset if you no longer need it. However, before you delete it, verify that no users in the organization plan to use it. You cannot retrieve projects, folders, or assets after you delete them.

You cannot delete an asset in the following situations:

- The asset is a model deployment that is currently available.
- The asset is used by another asset. You must first delete the dependencies of the asset before you can delete the asset.

For information about viewing asset dependencies, see ["Asset dependencies" on page 25](#).

Delete a project, folder, or asset from the **Explore** page, as shown in the following image:



1. On the **Explore** page, navigate to the object that you want to delete.

2. In the row that contains the project, folder, or asset, click **Actions** and select **Delete**.

Tags

A tag is an asset property that you can use to group assets. Create tags to filter for assets that share a common attribute on the **Explore** page.

For example, each of your organization's assets includes a tag that identifies the regional office that manages the asset. You want to view all of the assets that the Southwest regional office manages. On the **Explore** page, you explore by tag and then click the SW Region tag.

You can assign tags to all asset types. An asset can have up to 64 tags.

You can find all of the assets that have a particular tag using one of the following methods:

- Click the name of the tag in the **Tags** column, in any row.
- Explore by tag, and then in the list of tags that shows on the page, click the name of the tag.

Creating tags

Use an asset's **Properties** dialog box to create and assign tags for that asset or to create tags to be available for future use.

Perform the following steps to create multiple tags without assigning them to an asset:

1. On the **Explore** page, browse by asset type.
2. In a row that contains an asset, click **Actions** and select **Properties**.
3. In the **Tags** field, enter the name of a tag that you want to create, and then press Enter.

A tag can have a maximum of 255 characters.

You can't use the following characters in project, folder, asset, or tag names:

? ' | { } " ^ & [] / \

4. Continue to enter the desired tags. Press Enter after each tag name to add it to the tag list.

Properties: LinearRegressionDeployment



Name: LinearRegressionDeployment

Description: Model deployment for linear regression with one variable

Tags:

Classification ×

Linear_Regression ×



Save

Cancel

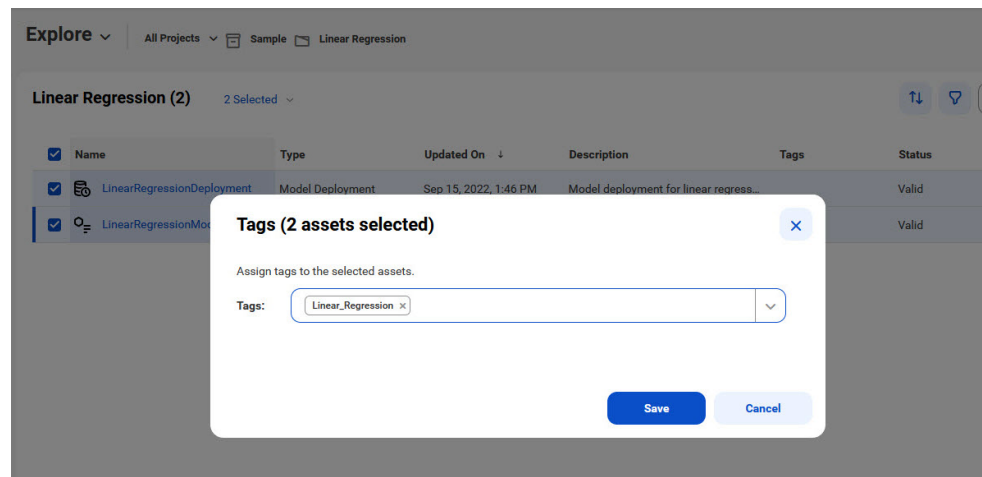
5. After you have entered the tags, delete the tags from the **Tags** field so that the asset does not become associated with the tags. The tags will still appear in the list of available tags.
6. Click **Save**.

Assigning tags

You can assign a tag to one asset at a time or assign a tag to multiple assets at the same time. You can also assign multiple tags to one asset.

When you assign tags to an asset, you can choose an existing tag or create a new one.

1. On the **Explore** page, navigate to the asset or assets.
2. Perform one of the following tasks depending on whether you want to assign tags to one asset or assign tags to multiple assets at the same time.
 - To assign tags to one asset, in the row that contains the asset, click **Actions** and select **Properties**.
 - To assign tags to multiple assets at the same time, in the row for each asset, select the check box. After you have selected all of the assets, from the Selection menu, select **Tags**.



3. Select an existing tag or enter the name of a new tag.
Continue adding tags or creating new tags until you have assigned all of the desired tags.
4. Click **Save**.

Editing and deleting tags

You can edit or delete a tag on the **Explore** page.

Edit a tag name or description in the tag properties. When you edit a tag, the properties for associated assets update as well. For example, if your `m_sales` asset has the `NorthWest` tag and you change the name of the tag to `NW`, the name of the tag changes to `NW` in the `m_sales` asset properties.

If you delete a tag, the tag no longer appears in the asset properties.

1. On the **Explore** page, browse by tags.
2. In the row that contains the tag, perform one of the following tasks:
 - To edit a tag, click **Actions** and select **Edit**. After you make your changes, click **Save**.
 - To delete a tag, click **Actions** and select **Delete**.

Asset dependencies

You can view object dependencies for an asset. You might want to view object dependencies before performing certain operations on an asset.

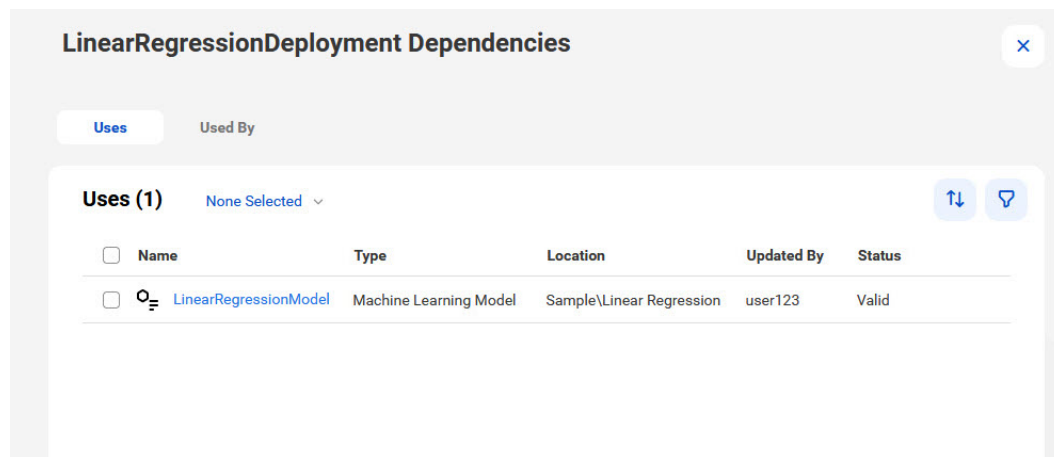
For example, you can't delete an asset if another object depends on the asset. You must first delete the dependent objects and then delete the asset. You can find the dependent objects by viewing the asset dependencies.

You can view object dependencies for an asset on the **Explore** page. To view object dependencies for an asset, in the row that contains the asset, click **Actions** and select **Show Dependencies**. The **Dependencies** page opens showing the **Uses** tab by default.

The **Uses** tab lists the objects that the selected asset uses.

The **Used By** tab lists the objects that use the selected asset.

The following image shows the **Dependencies** page for a model deployment with the **Uses** tab open:



To drill down to the lowest level dependency, you can continue to show dependencies for each asset that appears on the **Dependencies** page. At the top of the **Dependencies** page, a breadcrumb shows the chain of dependencies.

If you have the appropriate permissions, you can perform actions on the **Dependencies** page such as viewing or deleting assets. To view or delete an asset, in the row that contains the asset, click **Actions** and select the action.

Permissions

Permissions determine the access rights that a user has for a Secure Agent, Secure Agent group, connection, schedule, or asset. Permissions add additional or custom security for an object. Permissions define which users and groups can read, update, delete, execute, and change permissions on the object.

To configure permissions on an object, you need the following licenses and privileges:

- To configure permissions at the project level for all assets in a project, your organization must have the Set/Unset Security Permissions at Project Level license.

- To configure permissions at the folder level for all assets in a folder, your organization must have the Set/Unset Security Permissions at Folder Level license.
- To configure permissions on individual assets, your organization must have the Fine Grained Security license.
- The role assigned to your user account or to a group in which you are a member must have the Set Permission privilege for the object type. For example, to configure permissions on a Secure Agent, you must be assigned a role that has the Set Permission privilege for Secure Agents.

To configure permissions on an object, navigate to the object and set the appropriate permissions. For example, you want only users in the Development Team user group to have access to assets in the Development Data folder. Navigate to the folder, edit the permissions, and grant the Development Team user group permissions on the folder.

Permissions apply to the objects that you configure but not to copies of the object. Therefore, when you copy or export an asset, the permissions are not copied or exported with the asset. For example, you copy a model deployment in which only user rjones has execute permission. The copy of the deployment has no permissions assigned to it, therefore any user with privileges to execute model deployments can start the copy of the deployment.

You can configure the following permissions on an object:

Permission	Description
Read	Open and view the object.
Update	Edit the object. Requires read permission, which is automatically granted.
Delete	Delete the object.
Execute	Run the object. Applies to model deployments.
Change permissions	Change the permissions that are assigned to the object.

Note: These permissions control permissions within Informatica Intelligent Cloud Services. They do not control operating system permissions, such as the ability to start, stop, or configure the Secure Agent on Windows or Linux.

Rules and guidelines for permissions

Use the following rules and guidelines for permissions:

- When you configure permissions on an object, verify that the user or group to which you grant permissions is assigned a role with the appropriate privileges for the object type. For example, if you grant a user with the Monitor role Update privilege on a particular folder, the user cannot update the folder because the Monitor role does not have update privileges for folders.
- To edit an asset, the user must have read permission on all assets used within the asset. For example, when you assign a user Read and Update permissions on a model deployment, verify that the user also has Read permission on the machine learning model used in the deployment.

Configuring permissions

You can configure permissions on an object if you are assigned a role with the Set Permission privilege for the object type. For example, to configure permissions on a folder, you must be assigned a role that has the Set Permission privilege for folders.

1. Navigate to the object for which you want to configure permissions.

For example:

- To configure permissions on a machine learning model or model deployment, in Model Serve, open the project and folder that contain the asset.

2. In the row that contains the object, either click **Actions** and select **Permissions**, or click the **Change Permission** icon.

The **Permissions** dialog box lists the users and groups that have permissions on the object.

If the **Permissions** dialog box lists no users or groups, then no permissions are configured for the object. Any user with appropriate privileges for the object type can access the object.

3. To configure user permissions on the object:

- a. Select **Users**.
- b. If the user does not appear in the **Users** list, click **Add**, and select a user.
- c. Enable or disable the appropriate permissions on the user.

Note: When you grant any user permissions on the object, Informatica Intelligent Cloud Services also adds you as a user with permissions on the object. This prevents you from losing access to the object when you configure permissions.

4. To configure user group permissions on the object:

- a. Select **Groups**.
- b. If the group does not appear in the **Groups** list, click **Add**, and select a group.
- c. Enable or disable the appropriate permissions on the group.

Note: When you grant any group permissions on the object, Informatica Intelligent Cloud Services also adds you as a user with permissions on the object. This prevents you from losing access to the object when you configure permissions.

5. To remove all permissions restrictions for the object, remove all users and groups from the **Permissions** dialog box.

When you remove all users and groups, any user with appropriate privileges for the object type can access the object.

6. Click **Save**.

CHAPTER 6

Troubleshooting

A model file compressed with Apple's Archive Utility fails to upload.

When Archive Utility compresses a directory, it creates a directory named `__MACOSX` that includes files with more than one period in the file name. For security purposes, these files aren't permitted. Use a different application to compress the model file or exclude the `__MACOSX` directory when you compress the model file, and then upload the model file again.

Note: macOS hides the `__MACOSX` directory by default, so you might need to show hidden folders to view it.

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